CSC 3100: Data Structures

Midterm Exam (close book)

Time: 10:40-12:20 (100 mins), July 5, 2021

1. [20 marks] State and prove whether the following two statements are correct or not:

(1) $n^3 2^n + 6n^2 3^n = O(n^2 2^n)$; (2) $\frac{n^2}{\log n} = \Theta(n^2)$;

2. [15 marks] Develop an algorithm with pseudocodes for merging two sorted linked lists into one sorted linked list without duplicate data elements:

Input: *list1* and *list2*, each of which has been initialized with some nodes.

Output: *list3*, which (1) is a sorted linked list containing all the data elements from *list1* and *list2,* and (2) does not have any duplicate data element.

Assumption: (1) The head nodes of *list1* and *list2* can be accessed by *list1.head* and *list2.head* respectively; (2) for each node *x* in *list1* and *list2*, it has a data element and a next pointer, which can be accessed by *x.data* and *x.next*, respectively; (3) you can ignore the head nodes' data elements; and (4) all the data elements (which are integer values) in each list are sorted in an ascending order.

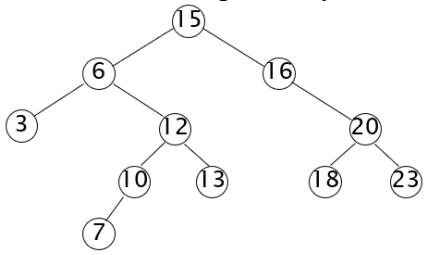
3. [15 marks] Design two stacks in one array A[*1…n*] ($n > 1$) in such a way that neither stack overflows unless the total number of elements in both stacks together is *n*. The PUSH and POP operations should run in *O(1)* time. Please briefly explain how to implement the two stacks, and then use pseudocodes to explain the steps of PUSH and POP for them.

4. [15 marks] Construct a binary tree such that its preorder=[8,5,1,7,10,12,15,11] and inorder = [1,5,7,8,10,15,12,11].

5. [15 marks] Insertion and deletion on binary search tree:

(1) [5 marks] Given an initial binary search tree as shown below, perform two consecutive operations: first delete key 15 from the tree, and then insert a new key 14 into the updated tree. Please draw these two updated trees after these operations respectively.

(2) [10 marks] Analyze the time complexities of key insertion and key deletion on binary search tree in the best case and worst case respectively.



6. [20 marks] Heap construction and HeapSort:

(1) [10 marks] Given an unsorted array A=[4,6,5,9,3,5], please use a sequence of tree graphs to illustrate how to build a max-heap.

(2) [10 marks] After building a max-heap on the array above, please use a sequence of tree graphs to illustrate how HeapSort(A) sorts the numbers in ascending order.